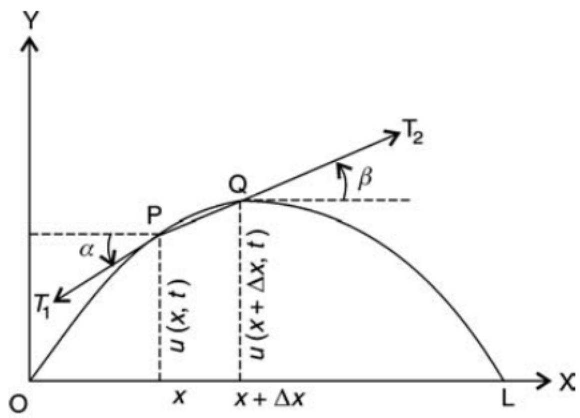
One-Dimensional Wave Equation

1.1 Derivation of One-Dimensional Wave Equation

Consider an elastic string, stretched to its length L and aligned (placed) along the $x-$axis, with its two ends $x = 0$ and $x = L$ fixed. Let ρ be the constant density of the string (i.e., homogeneous string). Let the function $u(x,t)$ denote the displacement of string at any point x and at any time $t > 0$ from the equilibrium position ($x-$axis). When the string is distorted, then it vibrates. The small transverse vibrations of such a vibrating string are mathematically modelled by one-dimensional wave equation.

Assume that the string is perfectly flexible and offers no resistance to bending. Applying Newton's second law of motion for a small portion of the string between x



and $x + \Delta x$, the equation is derived. Let T_1 and T_2 be tension at the end points P and Q of this portion of the string.

Assuming that points on the string move only in the vertical direction, there is no motion in the horizontal direction. Thus the sum of the forces in the horizontal direction must be zero *i.e.*, $-T_1 \cos\alpha + T_2 \cos\beta = 0$

$$\implies T_1 \cos\alpha = T_2 \cos\beta = T(\text{constant}) \tag{1.1}$$

Neglecting the gravitational force on the string, the only two forces acting on the string are the vertical components of tension $-T_1 \sin\alpha$ at P AND $T_2 \sin\beta$ at Q with upward direction takes as positive.

By Newton's second law *resultant of forces* = *mass* $*$ *acceleration*.Therefore,

$$T_2 \sin\beta - T_1 \sin\alpha = (\rho \Delta x)\left(\frac{\partial^2 u}{\partial t^2}\right) \tag{1.2}$$

Dividing (1.2) by (1.2), we have

$$\frac{T_2 \sin\beta}{T_2 \cos\beta} - \frac{T_2 \sin\alpha}{T_2 \cos\alpha} = \frac{\rho \Delta x}{T}\frac{\partial^2 u}{\partial t^2}$$

$$\tan\beta - \tan\alpha = \frac{\rho \Delta x}{T}\frac{\partial^2 u}{\partial t^2} \tag{1.3}$$

Replace $\tan\alpha$ by $\frac{\partial u(x,t)}{\partial x}$ and $\tan\beta$ by $\frac{\partial u(x+\Delta x,t)}{\partial x}$ because they are slopes of the string at x and $x + \Delta x$. Then

$$\frac{\partial u(x+\Delta x,t)}{\partial x} - \frac{\partial u(x,t)}{\partial x} = \frac{\rho \Delta x}{T}\frac{\partial^2 u}{\partial t^2}$$

Rewriting and taking the limit as $\Delta x \longrightarrow 0$

$$\lim_{\Delta x \longrightarrow 0} \frac{1}{\Delta x}\left[\frac{\partial u(x+\Delta x,t)}{\partial x} - \frac{\partial u(x,t)}{\partial x}\right] = \frac{\rho}{T}\frac{\partial^2 u}{\partial t^2}$$

Thus

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{1.4}$$

where $c^2 = \frac{T}{\rho}$. Equation (1.4) is known as one dimensional wave equation.

1.2 Solution of One-Dimensional Wave Equation by Separation of Variables

Consider an elastic string, placed along the $x-$axis, stretched to length L between two fixed points $x = 0$ and $x = L$. Let $u(x,t)$ denote the deflection (displacement from equilibrium position). Then the small transverse vibrations of the string is governed by the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{1.5}$$

where $c^2 = \frac{T}{\rho}$.

Since the string is fastened at the ends $x = 0$ and $x = L$, we have the two **boundary conditions**

$$(a)\ u(0,t) = 0 \qquad (b)\ u(L,t) = 0, \qquad \text{for all } t \geqslant 0 \tag{1.6}$$

Furthermore, the form of the motion of the string will depend on its initial deflection (deflection at time $t = 0$), call it $f(x)$, and on its initial velocity (velocity at $t = 0$), call it $g(x)$. We thus have the two **initial conditions**

$$(a)\ u(x,0) = f(x) \qquad (b)\ u_t(x,0) = g(x), \qquad (0 \leqslant x \geqslant L) \tag{1.7}$$

where $u_t = \frac{\partial u}{\partial t}$. We now have to find a solution of the PDE (1.5) satisfying the conditions (1.6) and (1.7). This will be the solution of our problem. We shall do this in three steps, as follows.

Step 1 : By the "method of separating variables", setting $u(x,t) = X(x)T(t)$, we obtain from (1.5) two ODEs, one for $X(x)$ and the other one for $T(t)$.

Step 2 : We determine solutions of these ODEs that satisfy the boundary conditions (1.6).

Step 3 : Finally, using Fourier series, we compose the solutions found in Step 2 to obtain a solution of (1.5) satisfying both (1.6) and (1.7), that is, the solution of our model of the vibrating string.

Step 1 Two ODEs from the Wave Equation

Assume that u is separable i.e.,

$$u(x,t) = X(x)T(t). \tag{1.8}$$

Differentiating 1.8, we obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 X}{\partial x^2}T \qquad \frac{\partial^2 u}{\partial t^2} = X\frac{\partial^2 T}{\partial t^2} \tag{1.9}$$

By substituting (1.9) in the wave equation (1.5) we have

$$X\frac{\partial^2 T}{\partial t^2} = c^2\frac{\partial^2 X}{\partial x^2}T \quad \Longrightarrow \quad \frac{1}{c^2 T}\frac{\partial^2 T}{\partial t^2} = \frac{1}{X}\frac{\partial^2 X}{\partial x^2}.$$

The variables are now separated, the left side depending only on t and the right side only on x. Hence both sides must be constant because, if they were variable, then changing t or x would affect only one side, leaving the other unaltered. Thus, say,

$$\frac{1}{c^2 T}\frac{\partial^2 T}{\partial t^2} = \frac{1}{X}\frac{\partial^2 X}{\partial x^2} = k$$

Multiplying by the denominators gives immediately two ordinary DEs

$$\frac{\partial^2 X}{\partial x^2} - kX = 0 \tag{1.10}$$

and

$$\frac{\partial^2 T}{\partial t^2} - c^2 kT = 0 \tag{1.11}$$

Step 2 Satisfying the Boundary Conditions

We now determine solutions X and T of (1.10) and (1.11) such that $u = XT$ satisfies the boundary conditions (1.6), that is,

$$(a)\ u(0,t) = X(0)T(t) = 0, \quad (b)\ u(L,t) = X(L)T(t) = 0 \quad \text{for all } t \tag{1.12}$$

We first solve (1.10). If $T \equiv 0$, then $u = XT \equiv 0$, which is of no interest. Hence, $T \not\equiv 0$ and then by (1.12),

$$(a)\ X(0) = 0, \quad (b)\ X(L) = 0 \tag{1.13}$$

We now show that k must be negative. It can be shown that for $k \geqslant 0$, only trivial solutions exist.

Let $k = 0$. Then by (1.10), $\frac{\partial^2 X}{\partial x^2} = 0 \implies \frac{\partial X}{\partial x} = a \implies X = ax + b$. From (1.13), when $x = 0$, $X = 0$ which gives $b = 0$. Also, when $x = L$, $X = 0$ which gives $a = 0$. Thus $X \equiv 0$ and $u = XT \equiv 0$, which is of no interest.

Let $k = \mu^2$, positive. Then by (1.10), $\frac{\partial^2 X}{\partial x^2} - \mu^2 X = 0$ and a general solution is given by $X = Ae^{\mu x} + Be^{-\mu x}$. Then from (1.13), we obtain $X \equiv 0$ as before.

So for non-trivial solutions consider $k = -p^2$. Then (1.10) becomes $\frac{\partial^2 X}{\partial x^2} + p^2 X = 0$ and has a general solution

$$X(x) = A\cos px + B\sin px.$$

Using (1.13)(a), $0 = X(0) = A.1 + B.0 \implies A = 0 \implies X(x) = B\sin px$.

Using (1.13)(b), $0 = X(L) = B\sin pL$. Since B should not be zero (if $B = 0$, it results in a trivial solution again),

$$\sin pL = 0 \implies pL = n\pi \implies p = \frac{n\pi}{L} \qquad (n \text{ is any integer})$$

Setting $B = 1$, we thus obtain infinitely many solutions $X(x) = X_n(x)$ where

$$X_n(x) = \sin\frac{n\pi}{L}x \qquad \text{for } n = 1, 2, 3, \cdots \tag{1.14}$$

(For negative integer n we obtain essentially the same solutions, except for a minus sign, because $\sin(-\alpha) = -\sin(\alpha)$.)

We now solve (1.11) with $k = -p^2 = -(\frac{n\pi}{L})^2$. Then (1.11) becomes $\frac{\partial^2 T}{\partial t^2} + \lambda_n^2 T = 0$ where $\lambda_n = cp = \frac{cn\pi}{L}$ and has a general solution

$$T_n(t) = A_n\cos\lambda_n t + B_n\sin\lambda_n t \tag{1.15}$$

Hence solutions of (1.5) satisfying (1.6) are $u_n(x,t) = X_n(x)T_n(t) = T_n(t)X_n(x)$, written out

$$u_n(x,t) = (A_n\cos\lambda_n t + B_n\sin\lambda_n t)\sin\frac{n\pi}{L}x \qquad (n = 1, 2, 3, \cdots) \tag{1.16}$$

These functions are called the eigen functions, or characteristic functions, and the values $\lambda_n = \frac{cn\pi}{L}$ are called the eigen values, or characteristic values, of the vibrating string. The set $\{\lambda_1, \lambda_2, \lambda_3, \cdots\}$ is called the spectrum.

Step 3 Solution of the Entire Problem. Fourier Series

The eigen functions (1.16) satisfy the wave equation (1.5) and the boundary conditions (1.6) (string fixed at the ends). A single u_n will generally not satisfy the initial conditions (1.7). But since the wave equation (1.5) is linear and homogeneous, it follows that the sum of finitely many solutions u_n is a solution of (1.5). To obtain a solution that also satisfies the initial conditions (1.7), we consider the infinite series

$$u(x,t) = \sum_{n=1}^{\infty} u_n(x,t) = \sum_{n=1}^{\infty}(A_n \cos \lambda_n t + B_n \sin \lambda_n t) \sin \frac{n\pi}{L} x. \tag{1.17}$$

The unknown constants A_n and B_n's are determined using the initial conditions.

Given Initial Displacement 1.7(a) : $u(x,0) = f(x)$

Put $t = 0$ in (1.17). Then

$$u(x,0) = \sum_{n=1}^{\infty} u_n(x,0) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x = f(x) \quad (0 \leqslant x \leqslant L). \tag{1.18}$$

Thus A_n's are the Fourier coefficients in the half range Fourier sine series expansion of $f(x)$ in the interval $(0, L)$. Hence

$$A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx \qquad (n = 1, 2, 3, \cdots) \tag{1.19}$$

Given Initial Velocity 1.7(b) : $u_t(x,0) = g(x)$

Differentiate (1.17) w.r.t. t

$$\frac{\partial u}{\partial t} == u_t(x,t) = \sum_{n=1}^{\infty} (-A_n\lambda_n \sin \lambda_n t + B_n\lambda_n \cos \lambda_n t) \sin \frac{n\pi}{L} x.$$

Put $t = 0$. Then

$$u_t(x,0) = \sum_{n=1}^{\infty} B_n\lambda_n \sin \frac{n\pi}{L} x = g(x). \tag{1.20}$$

Hence we choose the B_n's so that for $t = 0$ the derivative $\frac{\partial u}{\partial t}$ becomes the Fourier sine series of $g(x)$. Thus

$$\begin{aligned}
B_n\lambda_n &= \frac{2}{L}\int_0^L g(x)\sin\frac{n\pi x}{L}dx \\
B_n &= \frac{2}{\lambda_n L}\int_0^L g(x)\sin\frac{n\pi x}{L}dx \\
B_n &= \frac{2}{cn\pi}\int_0^L g(x)\sin\frac{n\pi x}{L}dx \quad \left(\text{where } \lambda_n = \frac{cn\pi}{L},\ n = 1, 2, 3, \cdots\right) \qquad (1.21)
\end{aligned}$$

Hence the general solution of the one-dimensional wave equation with boundary conditions (1.6) and initial conditions (1.7) is

$$u(x,t) = \sum_{n=1}^{\infty} u_n(x,t) = \sum_{n=1}^{\infty}\left[A_n\cos\left(\frac{cn\pi}{L}\right)t + B_n\sin\left(\frac{cn\pi}{L}\right)t\right]\sin\left(\frac{n\pi}{L}\right)x \qquad (1.22)$$

where A_n and B_n's are given by (1.19) and (1.21).

1.3 D'Alembert's Solution of the Wave Equation

Consider an elastic string, placed along the $x-$axis, stretched to length L between two fixed points $x = 0$ and $x = L$. Let $u(x,t)$ denote the deflection (displacement from equilibrium position). Then the small transverse vibrations of the string is governed by the one-dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} \qquad (1.23)$$

where $c^2 = \frac{T}{\rho}$.

Let $v = x + ct$ and $w = x - ct$ be two new independent variables. Then u becomes a function of v and w by the use of the chain rule. The derivatives in (1.23) can now be expressed in terms of derivatives with respect to v and w. Denoting partial derivatives by subscripts, we see that

$$v_x = 1, \; w_x = 1, \; v_t = c, \; w_t = -c$$

Let us denote $u(x,t)$ as a function of v and w, by the same letter u. Then

$$u_x = u_v v_x + u_w w_x = u_v + u_w$$

$$u_t = u_v v_t + u_w w_t = c(u_v - u_w)$$

We assume that all the partial derivatives involved are continuous, so $u_{wv} = u_{vw}$. Since $v_x = 1$ and $w_x = 1$, we obtain

$$u_{xx} = \frac{\partial u_x}{\partial x} = \frac{\partial}{\partial x}(u_v + u_w) = (u_v + u_w)_v v_x + (u_v + u_w)_w w_x = u_{vv} + 2u_{vw} + u_{ww}$$

$$u_{tt} = \frac{\partial}{\partial t}[c(u_v - u_w)] = c\,[(u_v - u_w)_v v_t + (u_v - u_w)_w w_t] = c^2(u_{vv} - 2u_{vw} + u_{ww})$$

By inserting these two results in (1.23) we get

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad &\Longrightarrow \quad c^2(u_{vv} - 2u_{vw} + u_{ww}) = c^2(u_{vv} + 2u_{vw} + u_{ww}) \\ &\Longrightarrow \quad 2u_{vw} = 0 \\ &\Longrightarrow \quad \frac{\partial^2 u}{\partial w \partial v} = 0 \end{aligned} \tag{1.24}$$

(1.24) can be readily solved by two successive integrations, first with respect to w and then with respect to v. This gives

$$\frac{\partial u}{\partial v} = h(v) \qquad \text{and} \qquad u = \int h(v)dv + \psi(w).$$

Here $h(v)$ and $\psi(w)$ are arbitrary functions of v and w, respectively. Since the integral is a function of v, say, $\phi(v)$, the solution is of the form $u = \phi(v) + \psi(w)$. In terms of x and t, we thus have

$$u(x,t) = \phi(x+ct) + \psi(x-ct). \tag{1.25}$$

This is known as the D'Alembert's solution of the wave equation (1.23).

D'Alembert's Solution Satisfying the Initial Conditions

$$(a)\ u(x,0) = f(x) \qquad (b)\ u_t(x,0) = g(x). \tag{1.26}$$

By differentiating (1.25), we have

$$u_t(x,t) = c\phi'(x+ct) - c\psi'(x-ct). \tag{1.27}$$

where primes denote derivatives with respect to the entire arguments $x+ct$ and $x-ct$ respectively. From (1.25), (1.26) and (1.27) we have

$$u(x,0) = \phi(x) + \psi(x) = f(x), \tag{1.28}$$

$$u_t(x,0) = c\phi'(x) - c\psi'(x) = g(x). \tag{1.29}$$

Now, divide (1.29) by c and

$$\begin{aligned} \int_{x_0}^{x} [\phi(s) - \psi(s)] ds &= \int_{x_0}^{x} \frac{1}{c} g(s) ds \\ \phi(x) - \psi(x) &= k(x_0) + \frac{1}{c} \int_{x_0}^{x} g(s) ds \end{aligned} \tag{1.30}$$

where $k(x_0) = \phi(x_0) - \psi(x_0)$.

If we add this to (1.28), then

$$\phi(x) = \frac{1}{2} f(x) + \frac{1}{2c} \int_{x_0}^{x} g(s) ds + \frac{1}{2} k(x_0). \tag{1.31}$$

Similarly, subtraction of (1.30) from (1.28) gives

$$\psi(x) = \frac{1}{2}f(x) - \frac{1}{2c}\int_{x_0}^{x} g(s)ds - \frac{1}{2}k(x_0). \tag{1.32}$$

Replacing x by $x + ct$ in (1.31), we get

$$\phi(x+ct) = \frac{1}{2}f(x+ct) + \frac{1}{2c}\int_{x_0}^{x+ct} g(s)ds + \frac{1}{2}k(x_0) \tag{1.33}$$

Replacing x by $x - ct$ in (1.32), we get

$$\psi(x-ct) = \frac{1}{2}f(x+ct) + \frac{1}{2c}\int_{x-ct}^{x_0} g(s)ds - \frac{1}{2}k(x_0) \tag{1.34}$$

Adding (1.33) and (1.34) we get

$$u(x,t) = \frac{1}{2}\left[f(x+ct) + f(x-ct)\right] + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)ds. \tag{1.35}$$

If the initial velocity is zero, we see that this reduces to

$$u(x,t) = \frac{1}{2}\left[f(x+ct) + f(x-ct)\right]. \tag{1.36}$$